

Pre-work · Vessels and Nerves of the Upper Limb

Vessels and Nerves of the Thorax and Upper Limb. Read the Part A chapter first. These are routes, not lists, so trace them with your finger before you try to name anything.

Module 3 · 4 of 5

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first.
This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the vessel and nerve chapter into mini visuals. Six chunks. Chains for the routes, grids for the nerves.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient fractures the shaft of the humerus and afterwards cannot extend the wrist or the fingers, though elbow flexion is intact. Name the nerve and explain the pattern.

Answer. The radial nerve.

Reasoning. Start with what is lost, not with the fracture. Wrist and finger extension is the job of the posterior compartment of the forearm, and one nerve supplies that whole compartment. Elbow flexion is intact, so the musculocutaneous nerve in the anterior compartment of the arm is fine, which rules it out. Now the fracture site confirms it: the radial nerve spirals around the shaft of the humerus in the radial groove, pressed directly against the bone, so a mid-shaft break and this nerve are in the same place. Compartment gives you the nerve, and the bone tells you where it was injured.

takes longer than three minutes you are copying instead of organizing.

1 CHAIN

Aortic arch to the hand

One chain with arrows, from the arch through subclavian, axillary, brachial, then splitting to radial and ulnar and joining as the palmar arches. Write the landmark at every name change, because the landmark is the answer.

2 SPLIT

Right side against left side, off the arch

Two branches. Show why the right arm's supply needs an extra vessel and the left arm's does not. Name all three branches of the arch on the correct side.

3 GRID

Superficial against deep veins

Vein down the side. System, course, and where it joins across the top. Eight rows. Star the one used for a blood draw.

4 LADDER

The brachial plexus, five levels

Five rungs, roots to branches. Write the count at each rung: five roots, three trunks, six divisions, three cords, five terminal branches. The counts are what tell you if you have missed one.

5 GRID

The five terminal nerves

Nerve down the side. Cord of origin, what it supplies, and the deficit its injury causes across the top. This grid answers most clinical questions on the chapter.

DRAWING 1

The arterial tree of the upper limb

BRAIN DUMP FIRST

Two minutes. Every named artery from the arch to the hand, in order.

Draw the trunk outline and one arm. Start at the aortic arch and draw all three branches, correctly asymmetrical. Follow the right side out: subclavian, axillary, brachial, then radial and ulnar, then both palmar arches. Mark the first rib, the lower border of teres major, and the cubital fossa as the three landmarks where names change. Add the deep brachial artery in the radial groove.



In your notebook, head the page **M3-1 The arterial tree of the upper limb** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name any artery on your drawing and say what it was called one segment earlier and what landmark changed it.

Set A - Name it

1. The subclavian artery becomes the axillary artery at the _____.
2. The vein most often used for drawing blood at the elbow is the _____.
3. The nerve supplying serratus anterior is the _____.
4. The cord that gives rise to the radial and axillary nerves is the _____.
5. Name the three branches of the aortic arch, in order.

Set B - Predict it

1. A shoulder dislocation damages a nerve at the surgical neck of the humerus. Name it and predict the deficit.

2. A patient knocks the medial epicondyle. Predict which fingers tingle and explain why.

3. The radial artery is used for an arterial line. Explain, anatomically, why the hand stays supplied.

4. A blood draw at the cubital fossa is taken too deep. Name the two structures immediately beneath the vein.

Set C - Tell it apart

1. Explain why there is one brachiocephalic artery but two brachiocephalic veins.

2. Distinguish the cephalic from the basilic vein by side and by where each ends.

6 HUB

The humerus and the nerves that hug it

One humerus in the middle. Spoke to axillary at the surgical neck, radial in the radial groove, ulnar behind the medial epicondyle, median in the cubital fossa. Write the injury beside each spoke.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

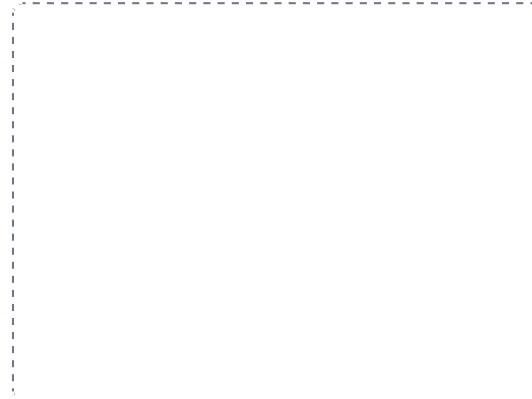
DRAWING 2

The brachial plexus, roots to branches

BRAIN DUMP FIRST

Two minutes. All five levels and the five terminal nerves.

Draw C5 through T1 as five lines on the left. Merge them into three trunks, split each trunk into two divisions, gather the divisions into three cords, and bring the cords to the five terminal nerves. Label every level. Then draw the axillary artery through the middle and show why the cords are named lateral, posterior, and medial.



In your notebook, head the page **M3-2 The brachial plexus, roots to branches** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can cover your labels and rebuild the whole plexus from the counts alone: five, three, six, three, five.

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3. Compare a median nerve injury at the wrist with an ulnar nerve injury at the elbow, by muscles lost and by hand appearance.
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YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

A cross section of the arm

BRAIN DUMP FIRST

One minute. Both compartments, with their vessels and nerves in place.

Draw the arm cut across at mid-shaft. Put the humerus in the middle and divide the surrounding tissue into anterior and posterior compartments with the intermuscular septa. Fill in the muscles of each. Then place the brachial artery and median nerve in the anterior compartment and the radial nerve with the deep brachial artery in the radial groove against the bone.



In your notebook, head the page **M3-3 A cross section of the arm** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain from your section why a mid-shaft humeral fracture produces wrist drop and not weak elbow flexion.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Aortic arch to the hand

CHAIN

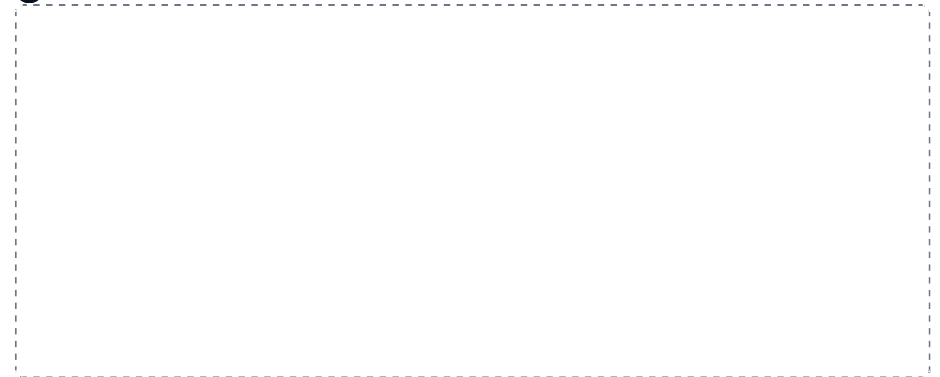


2 Right side against left side, off the arch

SPLIT

3 Superficial against deep veins

GRID



4 The brachial plexus, five levels

LADDER

5 The five terminal nerves

GRID

6 The humerus and the nerves that hug it

HUB